

$$② \int (\sin x + \cos x)^2 \cdot dx$$

$$\cos^2 x + \sin^2 x = 1$$

$$= \int (\sin^2 x + 2 \sin x \cos x + \cos^2 x) \cdot dx$$

$$\sin 2x = 2 \sin x \cos x$$

$$= \int (1 + 2 \sin x \cos x) \cdot dx$$

$$= \int (1 + \sin 2x) \cdot dx$$

$$= \int dx + \int \sin 2x \cdot dx$$

$$= x - \frac{1}{2} \cos(2x) + c$$

relations

.....

$$① \sin^2 x = \frac{1 - \cos 2x}{2}$$

$$② \cos^2 x = \frac{1 + \cos 2x}{2}$$

$$③ \sin 2x = 2 \sin x \cos x$$

$$④ \cos^2 x + \sin^2 x = 1$$

Integration of Trigonometric

$$1) \int \sin u \cdot du = -\cos u + C \quad \left\{ \begin{array}{l} \int \csc u \cdot \cot u \, du = -\csc u + C \\ \int \sec u \cdot \tan u \, du = \sec u + C \end{array} \right.$$

$$2) \int \cos u \, du = \sin u + C$$

$$3) \int \sec^2 u \cdot du = \tan u + C$$

$$4) \int \sec u \cdot \tan u \, du = \sec u + C$$

$$5) \int \csc^2 u \, du = -\cot u + C$$

Ex:

$$1) \frac{1}{2} \int \sec^2(2x) \, dx$$

$$\frac{1}{2} \tan(2x) + C$$

$$2) \frac{1}{4} \int \sin(2x) \, dx$$

$$-\frac{1}{8} \cos(2x) + C$$

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$$\cancel{f(x) = \frac{f(x+h) - f(x)}{h}}$$

$$\cancel{f(x) = \dots}$$

$$\bar{f}(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\bar{f}(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 3(x+h) + 2 - [x^2 + 3x + 2]}{h}$$

$$\bar{f}(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 3x + 3h + 2 - x^2 - 3x - 2}{h}$$

$$\bar{f}(x) = \lim_{h \rightarrow 0} \frac{2xh + h^2 + 3h}{h}$$

$$\bar{f}(x) = \lim_{h \rightarrow 0} \frac{h(2x + h + 3)}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h + 3$$

$$= \lim_{h \rightarrow 0} 2x + 0 + 3$$

$$= \lim_{h \rightarrow 0} 2x + 3$$



Final Exam of *Mathematic for information*
University of Basra
College of Computer Science and Information Technology
Department of Computer Information Systems



Exam Duration: 2 Hours

"B"

Date: 7 / 2/2023

Q1) Find the derivative y' , y'' of the following . (20M)

1) $y = \tan^{-1}(\cos(\sqrt{x}))$

2) $y = \tanh(5x^2 + 3)$

3) $y = \ln(\cos^{-1}(x))$

4) $y = e^{\sin^{-1}(x)}$

Q2) Answer the following (10M)

1. Let $y = f(x) + g(x)$ prove that $y' = f'(x) + g'(x)$.

2. Let $y = \tan(z)$ prove that $\frac{dy}{dx} = \sec^2(z) \cdot \frac{dz}{dx}$

Q3) Find equation for the tangent to the curve $y = x^2$ at the $x = -\frac{1}{2}$ (5M)

Good Luck

Examiner

Zahraa S. Bloshi



Final Exam of Business Statistical
University of Basra
College of Computer Science and Information Techn
Department of Computer Information Systems

Exam Duration: 1 Hours

Q1) Find the derivative y' of the following . (15m)

1_ $y = e^{\tan^{-1}(5x^2)}$

2_ $y = e^{\sin^{-1}(x^2)}$

3_ $y = 2^{\sinh(x^2)}$

4_ $y = e^{(\sin x + \sqrt[3]{x})}$

5_ $y = 9^{\frac{x}{3}}$

Q2) Find the integration of the following (25M)

1_ $\int \frac{\sin(5x)}{\cos^4(5x)} dx$

2_ $\int \frac{x^5}{(x+1)^7} dx$

3_ $\int \frac{x^3 + 1}{x^2 - x + 1} dx$

4_ $\int (\sin x + \cos x)^2 dx$

5_ $\int \frac{1 + \tan^2(x)}{\tan^3(x)} dx$

Good Luck